Australian Government
Geoscience Australia

AUSPOS GPS Processing Report

April 14, 2011

This document is a report of the GPS data processing undertaken by the AUSPOS Online GPS Processing Service (version: AUSPOS 2.01) . The AUSPOS Online GPS Processing Service uses International GNSS Service (IGS) products (final, rapid, ultra-rapid depending on availability) to compute precise coordinates in ITRF anywhere on Earth and GDA94 within Australia. The Service is designed to process only dual frequency GPS phase data.

An overview of the GPS processing strategy is included in this report.

Please direct any correspondence to geodesy@ga.gov.au

National Geospatial Reference Systems
Geoscience Australia
Cnr Jerrabomberra and Hindmarsh Drive
GPO Box 378, Canberra, ACT 2601, Australia
Freecall (Within Australia): 1800 800 173
Tel: +61 2 6249 9111. Fax +61 2 6249 9929
Geoscience Australia
Home Page: http://www.ga.gov.au

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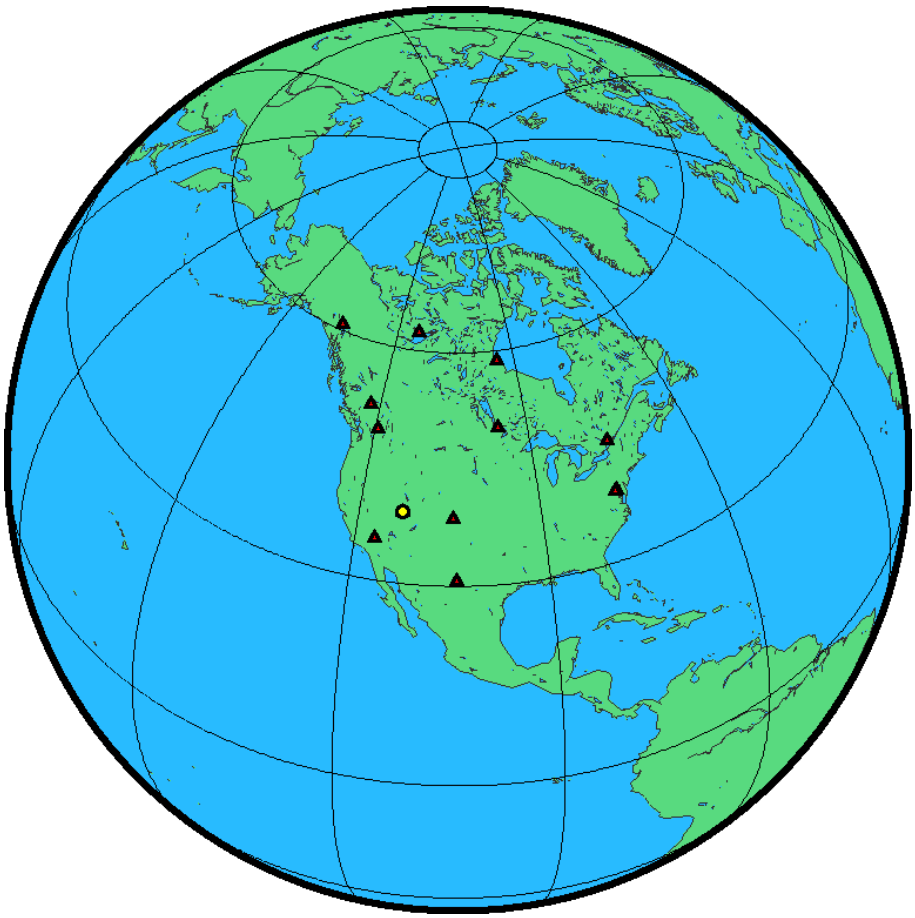
1 User Data

All antenna heights refer to the vertical distance from the Ground Mark to the Antenna Reference Point (ARP).

Station (s)	Submitted File	Antenna Type	Antenna Height (m)	Start Time	End Time
CLF1	CLF1.11o	TRM41249.00 TZGD	1.256	2011/03/24 16:14:30	2011/03/24 23:33:00

2 Processing Summary



Date	User Stations	Reference Stations	Orbit Type
2011/03/24 16:14:30	CLF1	AMC2 CHUR DRAO DUBO GODE GOLD MDO1 NRC1 USNO WHIT WILL YELL	IGS final

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3 Computed Coordinates, ITRF2005

All computed coordinates are based on the IGS realisation of the ITRF2005 reference frame. All the given ITRF2005 coordinates refer to a mean epoch of the site observation data. All coordinates refer to the Ground Mark.

3.1 Cartesian, ITRF2005

Station	X (m)	Y (m)	Z (m)	ITRF2005 @
CLF1	-1924370.116	-4553662.215	4018737.272	24/03/2011
AMC2	-1248596.289	-4819428.217	3976505.975	24/03/2011
CHUR	-236438.978	-3307616.887	5430049.277	24/03/2011
DRAO	-2059164.906	-3621108.413	4814432.306	24/03/2011
DUBO	-417603.805	-4064529.821	4881432.160	24/03/2011
GODE	1130773.652	-4831253.588	3994200.455	24/03/2011
GOLD	-2353614.427	-4641385.304	3676976.419	24/03/2011
MDO1	-1329998.860	-5328393.396	3236504.143	24/03/2011
NRC1	1112777.095	-4341475.873	4522955.851	24/03/2011
USNO	1112189.677	-4842955.041	3985352.294	24/03/2011
WHIT	-2218338.051	-2201205.120	5543057.519	24/03/2011
WILL	-2084258.258	-3313873.008	5019853.031	24/03/2011
YELL	-1224452.794	-2689216.167	5633638.285	24/03/2011

3.2 Geodetic, GRS80 Ellipsoid, ITRF2005

Geoid-ellipsoidal separations, in this section, are computed using a spherical harmonic synthesis of the global EGM2008 geoid. More information on the EGM2008 geoid can be found at http://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/

Station	Latitude (DMS)	Longitude (DMS)	Ellipsoidal Height(m)	Derived Above Geoid Height(m)
CLF1	39 17 48.44754	-112 54 31.82007	1370.051	1389.953
AMC2	38 48 11.24686	-104 31 28.54149	1911.405	1930.151
CHUR	58 45 32.67937	-94 05 19.42731	-19.283	28.882
DRAO	49 19 21.42620	-119 37 29.94017	541.899	558.447
DUBO	50 15 31.71168	-95 51 58.25564	245.288	274.981
GODE	39 01 18.22027	-76 49 36.59251	14.519	47.784
GOLD	35 25 30.56096	-116 53 21.30461	986.675	1017.367
MDO1	30 40 49.83913	-104 00 53.97785	2004.523	2026.602
NRC1	45 27 14.98602	-75 37 25.78557	82.518	116.050
USNO	38 55 08.26688	-77 03 58.41419	48.881	81.748
WHIT	60 45 01.84011	-135 13 19.60818	1427.411	1419.596
WILL	52 14 12.72201	-122 10 04.12769	1095.712	1110.456
YELL	62 28 51.21573	-114 28 50.53304	180.945	207.660

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4 Solution Information

4.1 Coordinate Precision - Geodetic, One Sigma

Station	σ East (m)	σ North (m)	σ Up (m)
CLF1	0.002	0.002	0.004
AMC2	0.002	0.002	0.003
CHUR	0.002	0.002	0.003
DRAO	0.002	0.002	0.003
DUBO	0.002	0.002	0.003
GODE	0.002	0.002	0.004
GOLD	0.002	0.002	0.003
MDO1	0.002	0.002	0.004
NRC1	0.002	0.002	0.003
USNO	0.002	0.002	0.004
WHIT	0.002	0.002	0.003
WILL	0.002	0.002	0.003
YELL	0.002	0.002	0.003

4.2 Ambiguity Resolution - per baseline

Baseline	Ambiguities Resolved	Baseline Length (km)
DRAO - WILL	87.0 %	370.4
DRAO - GOLD	80.0 %	1556.1
CHUR - YELL	83.3 %	1183.2
CHUR - DUBO	95.5 %	952.2
WHIT - YELL	76.7 %	1110.9
CLF1 - GOLD	84.2 %	555.6
AMC2 - GOLD	80.0 %	1158.7
WILL - YELL	87.5 %	1227.3
DUBO - NRC1	90.9 %	1596.0
AMC2 - MDO1	90.0 %	901.8
GODE - USNO	89.5 %	23.7
GODE - NRC1	89.5 %	721.0
AVERAGE	85.9%	946.4

Please note for a regional solution, such as used by AUSPOS, an average ambiguity resolution of 50% or better for the network indicates a reliable solution.

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5 Computation Standards

5.1 Computation System

Software	Bernese GPS Software Version 5.0.
GNSS system(s)	GPS only.

5.2 Data Preprocessing and Measurement Modelling

Data preprocessing	Phase preprocessing is undertaken in a baseline by baseline mode using triple-differences. In most cases, cycle slips are fixed by the simultaneous analysis of different linear combinations of L1 and L2. If a cycle slip cannot be fixed reliably, bad data points are removed or new ambiguities are set up A data screening step on the basis of weighted postfit residuals is also performed, and outliers are removed.
Basic observable	Carrier phase with an elevation angle cutoff of 10° and a sampling rate of 3 minutes. However, data cleaning is performed a sampling rate of 30 seconds. Elevation dependent weighting is applied according to $1/\sin(e)^2$ where e is the satellite elevation. The code observable is only used for the receiver clock synchronisation.
Modelled observable	Double differences of the ionosphere-free linear combination.
Ground antenna phase centre calibrations	IGS05 absolute phase-centre variation model is applied.
Tropospheric Model	A priori model is the Saastamoinen-based hydrostatic mapped with the dry-Niell.
Tropospheric Estimation	Zenith delay corrections are estimated relying on the wet-Niell mapping function in intervals of 2 hour. N-S and E-W horizontal delay parameters are solved for every 24 hours.
Tropospheric Mapping Function	Niell
Ionosphere	First-order effect eliminated by forming the ionosphere-free linear combination of L1 and L2.
Tidal displacements	Solid earth tidal displacements are derived from the complete model from the IERS Conventions 2003, but ocean tide loading is not applied.
Atmospheric loading	Not applied
Satellite centre of mass correction	IGS05 phase-centre variation model applied
Satellite phase centre calibration	IGS05 phase-centre variation model applied
Satellite trajectories	Best available IGS products.
Earth Orientation	Best available IGS products.

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5.3 Estimation Process

Adjustment	Weighted least-squares algorithm.
Station coordinates	Coordinate constraints are applied at the Reference sites with standard deviation of 5mm and 10mm for horizontal and vertical components respectively.
Troposphere	Zenith delay parameters and pairs of horizontal delay gradient parameters are estimated for each station in intervals of 2 hour and 24 hours.
Ionospheric correction	An ionospheric map derived from the contributing reference stations is used to aid ambiguity resolution using the QIF strategy
Ambiguity	Ambiguities are resolved in a baseline-by-baseline mode using Quasi-Ionosphere-Free (QIF) approach.

5.4 Reference Frame

Terrestrial reference frame	IGS05 station coordinates and velocities mapped to the mean epoch of observation.
Australian datum	GDA94 coordinates determined via Helmert transformation from ITRF using the Dawson and Woods (2010) parameters.
Derived AHD	For stations within Australia, AUSGeoid09 is used to compute AHD. AUSGeoid09 is the Australia-wide gravimetric quasigeoid model that has been a posteriori fitted to the Australian Height Datum
Above-geoid heights	Earth Gravitational Model EGM2008 released by the National Geospatial-Intelligence Agency (NGA) EGM Development Team is used to compute above-geoid heights. This gravitational model is complete to spherical harmonic degree and order 2159, and contains additional coefficients extending to degree 2190 and order 2159.

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